Experiment 5-Max–Min Composition of Fuzzy Relations

Objective

Write a Program to create a fuzzy relation through the Cartesian product of two fuzzy sets and perform max-min composition on fuzzy relations

Theory

A **fuzzy relation** is a fuzzy set defined on the Cartesian product of two or more universes, where each ordered pair is associated with a membership value in the range $[0, 1]$.

Let $R(x, y)$ be a fuzzy relation between sets X and Y, and $S(y, z)$ be a fuzzy relation between sets Y and Z. The **Max–Min composition** of R and S produces a new fuzzy relation $T(x, z)$ defined as:

$$T(x, z) = \max_y \left[\min\left(R(x, y), S(y, z)\right)\right]$$

This method selects the strongest possible connection between x and z through the weakest intermediate link.

Code

```
import numpy as np

R = np.array([
    [0.7, 0.3],
    [0.2, 0.9]
])

S = np.array([
    [1.0, 0.8, 0.3],
    [0.8, 0.4, 0.9]
])

T = np.zeros((R.shape[0], S.shape[1]))

for i in range(R.shape[0]):
    for j in range(S.shape[1]):
        T[i, j] = max(min(R[i, k], S[k, j]) for k in range(R.shape[1]))

print("Max--Min Composition T:")
print(T)
```

Output

```
Max--Min Composition T:
[[0.6 0.5 0.3]
 [0.8 0.4 0.7]]
```

Result

$$T = \begin{bmatrix} 0.6 & 0.5 & 0.3 \\ 0.8 & 0.4 & 0.7 \end{bmatrix}$$

Conclusion

The Max–Min composition of the given fuzzy relations R and S results in a 2×3 fuzzy relation T, representing the indirect relationship between the elements of the first and third universes.